Master Marine Physics, 2nd year
Geophysical Fluid Dynamics
Exercise sheet # 2 : Rob's question, then Supplement from Furevik §5 Coriolis

1 Coriolis parameter and Earth's rotation : What is a day ?

The rotation rate is $\Omega = 7.2921 \times 10^{-5}$ radians s^{-1}.

If you assume that the Earth makes one rotation per day, and 2π radians per rotation, what value of Ω do you find ? Do your calculation to at least 3 or 4 significant digits. If it's not precisely $\Omega = 7.2921 \times 10^{-5}$ radians s^{-1}, what went wrong ?

5 Coriolis force

In this exercise we investigate the relations between acceleration in a rotating coordinate system and acceleration in a fixed system. In particular, we will look at the two terms known as the centripetal and Coriolis accelerations.

We let $\mathbf{\Omega}$ be the vector defining the rotation of the Earth. Thus the magnitude of the vector denotes the angular velocity and the direction is defined such that the rotation is clockwise when looking in the direction of the vector. A point P is defined by the position vector $\mathbf{x}_f$ relative to a fixed coordinate system (non-rotating), and by the position vector $\mathbf{x}_r$ relative to a rotating system (see Figure 4).

5a) Explain why we have the relationship between the velocities

$$\frac{D\mathbf{x}_f}{Dt} = \frac{D\mathbf{x}_r}{Dt} + \mathbf{\Omega} \times \mathbf{x}_r.$$

The mean radius of the Earth is 6371 km, and the rotation rate is $7.292 \times 10^{-5}\, s^{-1}$. What is the speed of a hard working, still-sitting student in Bergen (60.40°N), relative to a fixed coordinate system ?

5b) Explain why we have the relationship between the accelerations

$$\frac{D^2\mathbf{x}_f}{Dt^2} = \frac{D^2\mathbf{x}_r}{Dt^2} + 2\mathbf{\Omega} \times \frac{D\mathbf{x}_r}{Dt} + \mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{x}_r).$$

What is the physical interpretation of each term ? What will be the acceleration for the same student as above ? How large is this compared to the gravitational acceleration ?

5c) Show that for a local coordinate system, having the x, y, z axes pointing respectively east, north and upward, the Coriolis term becomes

$$2\mathbf{\Omega} \times \frac{D\mathbf{x}_r}{Dt} = (2\Omega \cos\phi\, w - 2\Omega \sin\phi\, v)\,\mathbf{i} + 2\Omega \sin\phi\, u\,\mathbf{j} - 2\Omega \cos\phi\, u\,\mathbf{k},$$

where $(\mathbf{i}, \mathbf{j}, \mathbf{k})$ are the unit vectors and (u, v, w) the velocities along the three axes of direction, and ϕ is the latitude.

5d) Assume that a cannonball is shot straight west from Bergen at an average horizontal speed of 200 $\mathrm{m\,s^{-1}}$. After 30 s the cannonball hits the water. Neglecting friction, how far from the latitude of Bergen does the cannonball hit the water ? (Suggestion : Assume constant u, and calculate the acceleration and the distance travelled in the north-south direction during the flight.) [Did the Coriolis force help keep the ball up or push it down ? Can you find the velocity that would keep the ball up forever ? Does this make physical sense ? !]

5e) The Sotra Bridge outside Bergen (same latitude as Bergen) has a height of 30 m. If you drop a coin from the bridge, how far from the vertical line will the coin hit the water ? [You can replace this question with the Pont Albert Louppe outside Brest, between Le Relecq and Plougastel, which is also 30 m above the water, and is at latitude about 48.39°N.]